

Chemistry for Land Scientists

Science

May, 2026

Chemistry for Land Scientists (A10631)

Short Title: Chemistry for Land Scientists
Faculty Science and Computing
Department Science
Cluster Chemistry
Author EOWENS
Credits 5

Level: Introductory

Description of Module / Aims

This is a basic chemistry course dealing with the general principles and practical techniques of chemistry for land scientists (Forestry, Agriculture and Horticulture). No previous knowledge of chemistry is assumed. This is a very applied module with a strong emphasis on laboratory-based teaching.

Programmes

CHEM-0005	BSc in Agriculture (WD_SAGUL_D)	1 / 2 / M
CHEM-0005	BSc in Agriculture (WD_SAGUL_D)	2 / 4 / E
CHEM-0005	BSc in Forestry (WD_SFORS_D)	1 / 2 / M
CHEM-0005	BSc in Forestry (WD_SFORS_D)	2 / 4 / E
CHEM-0005	BSc in Horticulture (WD_SHORT_D)	1 / 2 / M
CHEM-0005	BSc in Horticulture (WD_SHORT_D)	2 / 4 / E
CHEM-0005	BSc in Horticulture (WD_SHORT_DX)	1 / 2 / M
CHEM-0005	BSc in Horticulture (WD_SHORT_DX)	2 / 4 / E

Indicative Content

- Matter, atomic structure and the periodic table
- Chemical reactions, the mole concept and stoichiometry
- Electronic structure of an atom and bonding
- Chemistry of water including H bonding solvent ability, surface tension, osmosis and conductivity
- Acid and bases including definitions of acids and bases, neutralisation reactions, pH, buffers and buffering capacity, role of pH and alkalinity on water and soil quality
- Endothermic and exothermic reactions

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the nuclear atom.
2. Calculate reaction yields using balanced chemical reactions.
3. Distinguish between ionic and covalent bonding, and outline intermolecular bonding and relate to solvent properties.
4. Describe the structure and properties of water.
5. Explain the difference between acidity and basicity, calculate theoretical pH and buffering capacity and their role in water and soil quality.
6. Communicate by written and verbal means the results of theory and practical-based exercises.
7. Demonstrate a keen awareness of the safe handling and disposal of chemicals and good laboratory practice (GLP).

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Tutorials.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
<i>Practical</i>	40%	6,7
<i>Tutorial/Problem Sheets</i>	10%	1,2,3,4,5

Assessment Criteria

- <40%:** No understanding of the basic principles of chemistry. Inadequate practical skills in the associated laboratories.
- 40%-49%:** Has a superficial knowledge of the concepts outlined in the indicative content. Has a basic level of competency and skills in the associated laboratories..
- 50%-59%:** Able to clearly describe concepts encountered in the indicative content. Displays good practical skills in the associated practicals.
- 60%-69%:** In addition to the above, able to apply problem-solving skills to new similar problems. Has a high level of skill in the associated laboratories.
- 70%-100%:** All the previous to an excellent level. Demonstrates an ability to put a numerical solution into a context and assess whether such solutions are meaningful. Engages in class discussions and has a high level of proficiency in the associated laboratories.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Practical	18	
Independent Learning	87	

Essential Material

- "Waterford Institute of Technology." Chemistry for Land Scientists. . www.moodle.ie

Supplementary Material

- Baird, C. and M. McCann. _Environmental Chemistry_. 5th ed. New York: W.H.Freeman and Co., 2012.

Requested Resources

- Room Type: Chemistry Lab